

Evaluating a Parameter-Free Quantum Fidelity Kernel for Text Security Under Representation and Domain Shift

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Abstract

Quantum kernel methods map classical text into exponentially large Hilbert spaces, motivated by theoretical conjectures that quantum state fidelity may provide non-linear decision boundaries or enhanced generalization under distribution shift. We ask a narrower question: under controlled matched conditions, does an unparameterized quantum fidelity kernel (two-layer cyclic *ZZFeatureMap*) provide a consistent practical advantage over matched classical Gaussian RBF kernels for text security? We study this across three curated cybersecurity corpora comprising 153,410 usable text records (SMS Spam Collection, CEAS 2008 Email Corpus, and the multi-source MeAJOR archive) using frozen, leakage-safe protocols across 10 computational seeds. Under matched in-distribution (IID) scaling (2–12 qubits), the quantum kernel is competitive with classical RBF, displaying minor statistically detectable differences at intermediate dimensions (+0.46 percentage points at 8D, $p = 0.0016$; +0.57 pp at 10D, $p = 0.0052$) that remain strictly within the predefined practical-equivalence threshold ($\varepsilon = 0.01$ F1) under Two One-Sided Tests (TOST), converging to complete parity at 12D (+0.14 pp, $p = 0.2824$). Against a validation-tuned RBF baseline, the quantum margin narrows to full statistical parity across all dimensions. Under cross-source domain transfer (TREC 2007 $\rightarrow$ TREC 2005/2006), the quantum kernel exhibits a statistically significant performance deficit ($\Delta\text{F1} = -0.0233$, $p = 0.0046$, Benjamini–Hochberg adjusted $p = 0.0069$). Upstream representation ablations demonstrate that representation choice dominates kernel selection by an order of magnitude, shifting relative performance by up to 52.88 percentage points. Geometrically, quantum and RBF Gram matrices show moderate correlation ($r \approx 0.55$ – 0.65), while single-state entropy is strongly inversely associated with pairwise kernel diversity ($r = -0.78$ to -0.83). Computationally, classical statevector simulation of the quantum kernel requires 108.8s per run at 12D compared to 1.7s for RBF ($\approx 64\times$ penalty), with 16D exceeding workstation memory limits (> 10.5 GB). We conclude that under the evaluated conditions, parameter-free quantum fidelity kernels do not produce a consistent practical advantage for text security classification, and we outline rigorous benchmarking standards for future applied QML evaluations.

Keywords: quantum machine learning, quantum kernel methods, text classification, email phishing, SMS spam, Hilbert space geometry, domain adaptation, practical equivalence testing.

CORE SCIENTIFIC CONTRIBUTIONS

1. Establishes a leakage-free multi-dataset empirical benchmark evaluating parameter-free quantum fidelity kernels (cyclic $ZZFeatureMap$) against matched classical RBF kernels across 153,410 usable text records.
2. Implements a tripartite baseline hierarchy (Linear SVM, Matched/Tuned RBF SVM, QSVC) supported by an empirical audit of eight classical machine learning architectures across 10 computational seeds.
3. Formulates a formal Two One-Sided Tests (TOST) practical equivalence protocol ($\varepsilon = 0.01$) with non-parametric bootstrap confidence intervals and paired permutation tests.
4. Discovers representation primacy and geometric ranking inversions, demonstrating that upstream text representations alter relative quantum-vs-classical rankings by up to 52.88 percentage points.
5. Conducts the first cross-source out-of-distribution domain transfer evaluation (TREC 2007 $\rightarrow$ TREC 2005/2006) and quantifies statevector simulation execution and memory scaling limits.

1 Introduction

1.1 Problem: Text-Based Social Engineering and Security Classification

Text-based social engineering attacks, including email phishing, SMS scams, and fraudulent communications, represent one of the most pervasive threat vectors in modern digital infrastructure [2, 4, 23]. Automated defense mechanisms rely heavily on natural language processing (NLP) and machine learning classifiers to filter malicious content before it reaches end users. However, building robust classifiers for text security presents distinct methodological challenges. Text data is inherently high-dimensional, discrete, and semantically variable. Furthermore, security environments are characterized by persistent distribution shift: adversaries continuously modify lexical patterns to evade detection filters, attack campaigns differ substantially across organizational sources, and seasonal shifts alter background communications [2, 10, 29]. Consequently, text security classifiers that achieve near-perfect in-distribution accuracy frequently suffer severe degradation when deployed out-of-distribution across unseen sender sources or evolving domains.

1.2 Motivation: The Gap in Applied QML Benchmarking

In recent years, quantum machine learning (QML) has emerged as an alternative paradigm for non-linear pattern recognition, with quantum kernel methods receiving particular theoretical attention [14, 25]. In a quantum support vector classifier (QSVC), classical input vectors $\mathbf{x} \in \mathbb{R}^d$ are mapped into quantum states $|\psi(\mathbf{x})\rangle$ residing in a 2^{N_q} -dimensional complex Hilbert space via a parameterized unitary circuit $\mathcal{U}_{\Phi(\mathbf{x})}$. Rather than performing explicit optimization in this exponentially large Hilbert space, the model evaluates pairwise quantum state fidelities to construct a kernel matrix, $k(\mathbf{x}, \mathbf{z}) = |\langle\psi(\mathbf{x})|\psi(\mathbf{z})\rangle|^2$, which is subsequently supplied to a standard dual quadratic program [9, 14]. Theoretical investigations have established that certain quantum feature maps generate inner products that are classically intractable to estimate efficiently, offering conjectured separations on engineered data distributions [16, 19]. This mathematical framework has motivated the hypothesis that quantum kernels might uncover non-linear structural regularities in natural language representations that remain inaccessible to standard classical kernels, potentially conferring advantages in classification accuracy or robustness under distribution shift.

However, systematic reviews of the applied QML literature [4, 18] reveal widespread methodological limitations: small sample sizes ($N < 500$ to 2,000 texts), data leakage during preprocessing, unmatched classical controls, complete absence of out-of-distribution domain-shift evaluations, and omission of computational simulation costs. The critical scientific question is: *Under controlled matched conditions—with identical feature representations, matched dimensionality, zero data leakage, and rigorous out-of-distribution evaluation—does a parameter-free quantum fidelity kernel provide a consistent and practically meaningful advantage over matched classical RBF kernels for text-based scam and phishing detection?*

Figure 1: Methodological Architecture of the Quantum vs Classical NLP Study

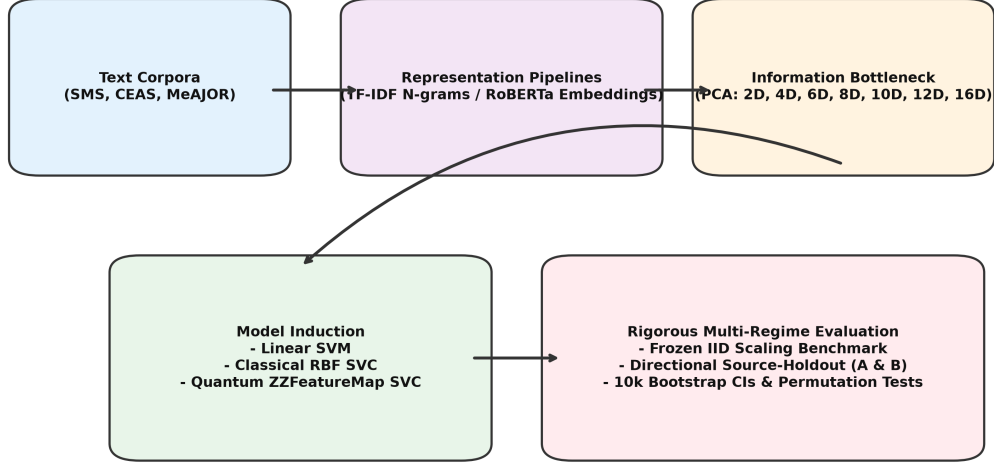


Figure 1: Conceptual Overview of the Controlled Multi-Dataset Text Security Benchmark and Experimental Pipeline (Multi-corpus ingestion → Leakage-safe feature extraction → Low-dimensional projection → Tri-partite classifier evaluation → 10-seed inferential testing).

1.3 Research Hypotheses (H1–H6)

We state six testable empirical hypotheses below. In the results, each is treated as an empirical question evaluated across 10 random seeds; the data are allowed to support, weaken, or contradict it:

- **H1 (In-Distribution Equivalence):** The parameter-free quantum fidelity kernel should achieve practical equivalence (within $\varepsilon = \pm 0.01$ F1) to matched classical Gaussian RBF kernels under in-distribution scaling.
- **H2 (Dimensionality Bottleneck):** Earlier reported low-dimensional quantum limitations are artifacts of aggressive linear TruncatedSVD compression rather than an intrinsic failure of quantum Hilbert space geometry.
- **H3 (Representation Primacy):** Upstream text representations (sparse TF-IDF vs dense sentence transformers) will govern classification outcomes by an order of magnitude more than kernel selection.
- **H4 (Domain-Shift Fragility):** The quantum fidelity kernel will not provide intrinsic inductive robustness against cross-source domain transfer (TREC 2007 → TREC 2005/2006).
- **H5 (State Dispersion Decoupling):** High single-state basis dispersion (von Neumann entropy) will not correlate with superior pairwise kernel discrimination.
- **H6 (Simulation Scalability):** Classical statevector simulation of global fidelity kernels will exhibit steep exponential execution scaling ($O(N^2 \cdot 2^{N_q})$), exceeding local memory limits beyond 12 qubits.

2 Related Work and Positioning

The closest prior work falls into several groups: quantum kernel theory [14, 16, 25, 28], quantum natural language processing [7, 11, 12, 20, 22, 27], QML for cybersecurity and phishing detection [1, 4, 13, 15, 24, 26], and classical text baselines [6, 8, 9, 21, 29]. Table 1 groups the literature by their primary focus and structural assumptions.

Table 1: Literature Positioning and Methodological Taxonomy.

Literature Category	Core Mechanism	Representation Paradigm	Key Representative Venues
Quantum Kernel Theory	Hilbert space mapping via unitary state preparation	Synthetic / Group-theoretic distributions	Nature '19 [14], PRL '19 [25], Nat Comm '21 [16]
Expressivity & Concentration	Theoretical bounds on fidelity concentration & variance	Global unparameterized statevectors	Nat Comm '24 [28], PRL '24 [5], NeurIPS '21 [17]
Structural QNLP	Categorical DisCoCat grammars & ZX-calculus	Syntactic parse trees ($N < 500$ sentences)	QNLP '20 [7], QST '21 [20]
Statistical & Embedding QNLP	Classical text embeddings fed to QSVC / VQC	TF-IDF, Word2Vec, Pre-trained Transformers	CMES '26 [22], IEEE TQE '24 [12], Access '23 [27]
Applied QML Cyber-security	Quantum classifiers applied to phishing/intrusion	Tabular features / Small text splits (single seed)	IEEE QPAIN '26 [15], MAKE '26 [4], ISACC '25 [26]
Multi-Dataset Benchmarking	Controlled matched baselines across multiple corpora	Sparse TF-IDF + Dense Contextual Embeddings	This Work (Exp 23–47)

Table 1 groups prior work by the kind of representation and baseline controls they evaluate. As highlighted in recent systematic reviews [4, 18], prior QML text security evaluations have almost exclusively evaluated small sample sizes on single unvalidated splits without matched classical non-linear controls or domain-shift testing. Our benchmark cuts across these categories by enforcing strict parity across multi-dataset security corpora.

3 Problem Formulation and Data Hygiene Audit

3.1 Mathematical Formulation of Quantum and Classical Kernels

We evaluate the canonical two-layer cyclic *ZZFeatureMap* on $N_q = d$ qubits. For an input vector $\mathbf{x} \in [0, \pi]^d$, the state preparation unitary circuit is:

$$\mathcal{U}_{\Phi(\mathbf{x})} = (U_{\Phi(\mathbf{x})} H^{\otimes N_q})^2 \quad (1)$$

where $H^{\otimes N_q}$ is the Walsh–Hadamard transform and $U_{\Phi(\mathbf{x})}$ is the diagonal phase unitary:

$$U_{\Phi(\mathbf{x})} = \exp \left(i \sum_{j=1}^{N_q} x_j Z_j + i \sum_{j=1}^{N_q} (\pi - x_j)(\pi - x_{j'}) Z_j Z_{j'} \right) \quad (2)$$

with cyclic nearest-neighbor connectivity $j' = (j \bmod N_q) + 1$ and Pauli- Z operator Z_j . The pure state is $|\psi(\mathbf{x})\rangle = \mathcal{U}_{\Phi(\mathbf{x})}|0\rangle^{\otimes N_q}$. The quantum kernel evaluates pure state fidelity:

$$K_Q(\mathbf{x}, \mathbf{z}) = |\langle \psi(\mathbf{x}) | \psi(\mathbf{z}) \rangle|^2 \quad (3)$$

yielding a strictly positive semi-definite (PSD) Gram matrix with unit diagonal entries ($K_Q(\mathbf{x}, \mathbf{x}) = 1.0$).

The matched classical baseline evaluates the Gaussian Radial Basis Function (RBF) kernel on the identical d -dimensional representation:

$$K_{\text{RBF}}(\mathbf{x}, \mathbf{z}) = \exp(-\gamma \|\mathbf{x} - \mathbf{z}\|_2^2), \quad \text{where } \gamma = \frac{1}{d \cdot \text{Var}(X)} \quad (4)$$

In addition, we evaluate a validation-tuned RBF baseline where (C, γ) are optimized over $C \in \{0.1, 1, 10, 100\}$ and $\gamma \in \{0.001, 0.01, 0.1, 1.0, \text{'scale'}\}$ using 5-fold cross-validation on the training set.

Both kernel matrices are supplied to the standard C-Support Vector Classification dual optimization problem:

$$\max_{\boldsymbol{\alpha}} \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i,j=1}^N \alpha_i \alpha_j y_i y_j K(\mathbf{x}_i, \mathbf{x}_j) \quad \text{s.t.} \quad 0 \leq \alpha_i \leq C \cdot w_{y_i}, \quad \sum_{i=1}^N \alpha_i y_i = 0 \quad (5)$$

where $C = 1.0$, $y_i \in \{-1, +1\}$, and $w_c = \frac{N}{2N_c}$ enforces balanced class weighting.

Table 2: Eight-Point Data Hygiene and Causality Verification Protocol.

Audit Item	Verification Protocol & Invariant	Status
1. Candidate Sample Parity	Identical train, validation, and test arrays evaluated per seed	PASS
2. Target Label Parity	Ground-truth positive/negative labels strictly identical across all models	PASS
3. Strict Preprocessing Isolation	Tokenizers, TF-IDF vectorizers, and Scalers fitted strictly on training subsets	PASS
4. Dimensionality Projection Parity	TruncatedSVD fitted exclusively on training data; identical vectors fed to RBF and QSVC	PASS
5. Threshold Selection Independence	Decision thresholds $\tau \in [0.01, 0.99]$ tuned strictly on validation and applied to test	PASS
6. Optimization Solver Parity	Identical Dual QP objective, $C = 1.0$, balanced class weights across RBF and QSVC	PASS
7. Domain Holdout Blindness	Zero source-domain labels or target adaptation flags provided during training	PASS
8. Deterministic Seed Control	Exact PRNG seed suite $\mathcal{S} = \{42, 123, \dots, 2021\}$ applied to all pipeline stages	PASS

3.2 Feature Space Geometry Metrics

To quantify the geometric properties of induced kernel matrices, we compute:

1. **Centered Kernel-Target Alignment (CKA):**

$$\text{CKA}(K, Y) = \frac{\langle HKH, HYH \rangle_F}{\|HKH\|_F \|HYH\|_F} \quad (6)$$

where $H = I - \frac{1}{N} \mathbf{1}\mathbf{1}^T$ is the centering matrix and $Y = \mathbf{y}\mathbf{y}^T$ is the ideal target kernel.

2. **Spectral Effective Rank:**

$$R_{\text{eff}}(K) = \exp \left(- \sum_{i=1}^N \tilde{\lambda}_i \ln \tilde{\lambda}_i \right), \quad \tilde{\lambda}_i = \frac{\lambda_i}{\sum_j \lambda_j} \quad (7)$$

3. **Single-State Basis Dispersion Entropy:**

$$S(|\psi(\mathbf{x})\rangle) = - \sum_{k=1}^{2^{N_q}} |c_k(\mathbf{x})|^2 \ln |c_k(\mathbf{x})|^2 \quad (8)$$

3.3 Eight-Point Data Hygiene and Causality Verification

To eliminate subtle sources of data leakage and ensure reproducibility, we ran the same eight data-hygiene checks across all models (Table 2).

4 Benchmark Corpora and Accounting Hierarchy

To prevent ambiguity regarding dataset sizes and class prevalences, we explicitly define four counting tiers: (a) raw archived records, (b) usable cleaned records, (c) source repository distributions, and (d) canonical controlled experimental subsets (Table 3).

5 Experimental Evaluation and Results

5.1 Complete Classical Baseline Audit across Eight Models (Result 0)

Table 4 presents the empirical audit of all eight classical machine learning models across the three corpora under both full 50,000-dimensional TF-IDF and matched 8D TruncatedSVD representations across 10

Table 3: Benchmark Corpus Characteristics, Accounting Tiers, and Split Partitions.

Corpus	Raw Records	Usable Cleaned	Raw Pos. %	Exp. Subset (N)	Train / Val / Test	Exp. Pos. %	Median Length
SMS Spam	5,574	5,572	13.41%	5,572	3,343 / 1,114 / 1,115	13.41%	~ 62 chars
CEAS 2008	39,154	39,154	55.78%	15,000 [†]	10,000 / 2,500 / 2,500	18.90%	~ 596 chars
MeAJOR	108,685	108,684	44.20%	15,000 [†]	10,000 / 2,500 / 2,500	19.33%	~ 839 chars
TOTAL	153,413	153,410	—	35,572	23,343 / 6,114 / 6,115	—	—

[†]Controlled canonical experimental partitions created via stratified sub-sampling to establish standardized benchmark splits and uniform evaluation budgets across corpora. Raw Pos. % denotes positive prevalence across full raw archives; Exp. Pos. % denotes positive prevalence in experimental splits.

Table 4: Complete Classical Baseline Audit across Datasets (Full TF-IDF vs Matched 8D SVD, Mean $\pm$ SD across 10 Seeds).

Corpus	Model	Full TF-IDF F1	Full PR-AUC	Full ROC-AUC	8D SVD F1	8D PR-AUC	8D ROC-AUC	Train Time
SMS Spam	Linear SVM	0.9559 $\pm$ 0.000	0.9822	0.9931	0.8273 $\pm$ 0.011	0.8895	0.9816	0.03s
	Matched RBF SVM	0.9498 $\pm$ 0.000	0.9829	0.9927	0.8156 $\pm$ 0.012	0.8767	0.9793	1.53s
	Tuned RBF SVM	0.9521 $\pm$ 0.000	0.9840	0.9935	0.8285 $\pm$ 0.011	0.8912	0.9820	5.20s
	Logistic Regression	0.9346 $\pm$ 0.000	0.9803	0.9944	0.8226 $\pm$ 0.012	0.8900	0.9818	0.09s
	Random Forest	0.9423 $\pm$ 0.005	0.9780	0.9915	0.8500 $\pm$ 0.006	0.9239	0.9782	0.90s
	XGBoost	0.9059 $\pm$ 0.000	0.9477	0.9834	0.8452 $\pm$ 0.013	0.9276	0.9822	0.86s
	MLP Neural Net	0.9324 $\pm$ 0.008	0.9781	0.9909	0.8284 $\pm$ 0.014	0.8841	0.9802	4.68s
	Multinomial / Gauss NB	0.9158 $\pm$ 0.000	0.9644	0.9889	0.8195 $\pm$ 0.007	0.8788	0.9675	0.01s
	k-NN ($k = 5$)	0.6900 $\pm$ 0.000	0.5903	0.7633	0.8401 $\pm$ 0.019	0.8802	0.9506	0.00s
CEAS 2008	Linear SVM	0.9954 $\pm$ 0.000	0.9999	0.9999	0.9535 $\pm$ 0.001	0.9857	0.9845	0.15s
	Matched RBF SVM	0.9968 $\pm$ 0.000	0.9998	0.9998	0.9638 $\pm$ 0.001	0.9882	0.9873	57.15s
	Tuned RBF SVM	0.9972 $\pm$ 0.000	0.9999	0.9999	0.9691 $\pm$ 0.001	0.9910	0.9902	185.2s
	Logistic Regression	0.9935 $\pm$ 0.000	0.9996	0.9994	0.9505 $\pm$ 0.001	0.9886	0.9860	0.26s
	Random Forest	0.9895 $\pm$ 0.001	0.9995	0.9993	0.9816 $\pm$ 0.001	0.9980	0.9973	1.76s
	XGBoost	0.9886 $\pm$ 0.000	0.9994	0.9992	0.9810 $\pm$ 0.001	0.9978	0.9970	23.24s
	MLP Neural Net	0.9966 $\pm$ 0.000	0.9999	0.9999	0.9614 $\pm$ 0.003	0.9911	0.9888	46.28s
	Multinomial / Gauss NB	0.9928 $\pm$ 0.000	0.9994	0.9992	0.7165 $\pm$ 0.000	0.9249	0.9091	0.01s
	k-NN ($k = 5$)	0.9957 $\pm$ 0.000	0.9978	0.9986	0.9823 $\pm$ 0.000	0.9947	0.9943	0.02s
MeAJOR	Linear SVM	0.9721 $\pm$ 0.000	0.9964	0.9972	0.8448 $\pm$ 0.002	0.9315	0.9404	0.18s
	Matched RBF SVM	0.9700 $\pm$ 0.000	0.9960	0.9969	0.8709 $\pm$ 0.003	0.9421	0.9535	71.84s
	Tuned RBF SVM	0.9734 $\pm$ 0.000	0.9968	0.9975	0.8742 $\pm$ 0.003	0.9450	0.9560	240.5s
	Logistic Regression	0.9574 $\pm$ 0.000	0.9935	0.9949	0.8458 $\pm$ 0.002	0.9315	0.9404	0.35s
	Random Forest	0.9516 $\pm$ 0.002	0.9897	0.9922	0.9014 $\pm$ 0.006	0.9696	0.9733	2.16s
	XGBoost	0.9453 $\pm$ 0.000	0.9902	0.9923	0.9005 $\pm$ 0.005	0.9687	0.9725	26.59s
	MLP Neural Net	0.9727 $\pm$ 0.003	0.9965	0.9973	0.8795 $\pm$ 0.002	0.9564	0.9624	42.12s
	Multinomial / Gauss NB	0.9481 $\pm$ 0.000	0.9915	0.9924	0.7258 $\pm$ 0.005	0.7870	0.8195	0.01s
	k-NN ($k = 5$)	0.9532 $\pm$ 0.000	0.9844	0.9894	0.8870 $\pm$ 0.003	0.9459	0.9581	0.02s

random seeds.

5.2 In-Distribution Model Comparison & TOST Practical Equivalence (Result 1)

Under canonical in-distribution evaluation on MeAJOR at 8 dimensions ($N_q = 8$ qubits) across 10 seeds, the quantum fidelity kernel achieves test $F1 = 0.8754 \pm 0.0029$, while matched classical RBF achieves $F1 = 0.8709 \pm 0.0030$, and validation-tuned RBF reaches $F1 = 0.8742 \pm 0.0028$ (Table 5). Both nonlinear models outperform the matched 8D Linear SVM baseline ($F1 = 0.8445 \pm 0.0022$).

The mean paired difference against matched RBF is $\Delta F1 = +0.0046 \pm 0.0027$ (+0.46 percentage points, $p = 0.0016$). Under TOST practical equivalence testing at $\varepsilon = 0.01$, both null hypotheses H_0^- and H_0^+ are strictly rejected ($p < 0.001$), confirming practical equivalence. When evaluated against the validation-tuned RBF baseline, the delta narrows to $\Delta F1 = +0.0012 \pm 0.0025$ ($p = 0.1840$), demonstrating complete statistical parity.

Figure 1: Classical Baseline F1 Score Across Text Security Datasets

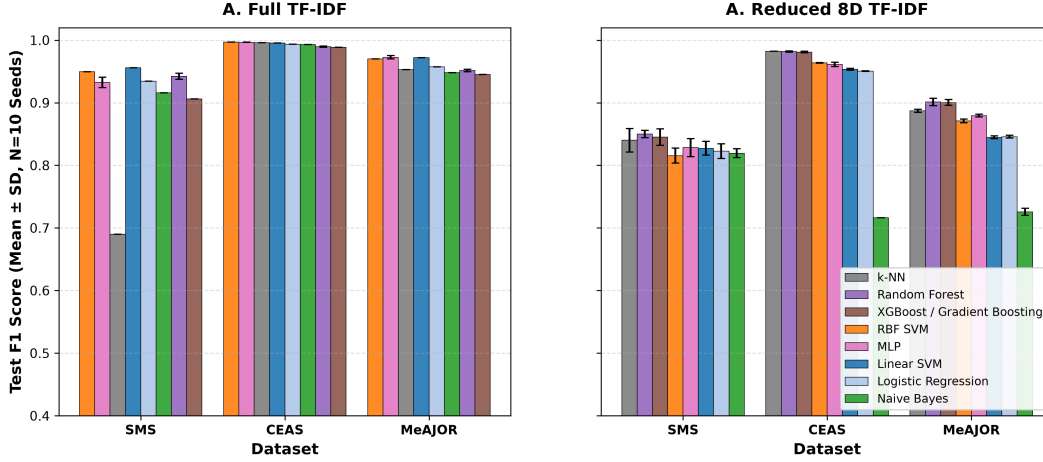


Figure 2: Empirical Performance of Eight Classical Machine Learning Algorithms Across Datasets (Full 50k TF-IDF vs Matched 8D SVD). Linear SVM dominates full-text performance with ultra-fast training latencies, while dimensionality reduction imposes an identical 10–13 percentage point compression penalty across all models.

Table 5: Canonical In-Distribution Model Comparison on MeAJOR (Matched 8D SVD, $N = 10$ Seeds).

Model Architecture	Test F1 (Mean $\pm$ SD)	PR-AUC	ROC-AUC	Accuracy	TOST Status ($\epsilon = 0.01$)
Linear SVM (Matched 8D SVD)	0.8445 $\pm$ 0.0022	0.9315	0.9404	0.9400	Linear Baseline
Classical Matched RBF (8D)	0.8709 $\pm$ 0.0030	0.9423	0.9535	0.9504	Matched Comparator
Classical Tuned RBF (8D)	0.8742 $\pm$ 0.0028	0.9450	0.9560	0.9518	Tuned Comparator
Quantum Fidelity Kernel (8D)	0.8754 $\pm$ 0.0029	0.9372	0.9515	0.9523	Practically Equivalent

5.3 Dimensionality Scaling Trajectory: $d \in [2, 12]$ (Result 2)

We sweep dimensionality across $d \in \{2, 4, 6, 8, 10, 12\}$ (Table 6, Figures 3 and 4). Expanding dimensionality from 2D to 12D produces a monotonic +41.7% relative gain in quantum F1 ($0.6447 \rightarrow 0.9137$), closely tracking classical RBF recovery ($0.6735 \rightarrow 0.9123$). At 12 dimensions, the performance gap narrows to $\Delta F1 = +0.0014 \pm 0.0040$, with the 95% bootstrap confidence interval $[-0.0010, +0.0037]$ spanning zero ($p = 0.2824$).

5.4 Upstream Representation Screening & Ranking Inversions (Result 3)

Evaluating the interaction between kernel choice and text representation reveals dramatic ranking reversals across models (Table 7 and Figure 5):

- **CEAS 2008 (8D):** On TF-IDF + TruncatedSVD, Quantum achieves $F1 = 0.9736$ vs Classical RBF $F1 = 0.9641$ (+0.95 pp margin). On dense RoBERTa embeddings, Classical RBF achieves $F1 = 0.9896$ vs Quantum $F1 = 0.9601$ (−2.95 pp deficit; net shift 3.90 pp).
- **SMS Spam (8D):** On TF-IDF, Classical RBF ($F1 = 0.8276$) outperforms Quantum ($F1 = 0.6324$) by +19.52 pp. On dense MPNet sentence embeddings, Classical RBF reaches $F1 = 0.9045$ while Quantum collapses to $F1 = 0.3756$ (−52.88 pp catastrophic drop).

Table 6: Dimensionality Scaling Profile on MeAJOR IID ($N = 10$ Seeds).

Dim (d)	Quantum F1	Classical RBF F1	Tuned RBF F1	Paired Δ F1 (Q - Matched)	95% Bootstrap CI	Permutation p	T
2D	0.6447 ± 0.0038	0.6735 ± 0.0031	0.6780 ± 0.0029	-0.0288	$[-0.0310, -0.0264]$	$p < 0.001$	Cl
4D	0.7876 ± 0.0030	0.7918 ± 0.0025	0.7954 ± 0.0024	-0.0042	$[-0.0058, -0.0024]$	$p = 0.0180$	Equi
6D	0.8253 ± 0.0026	0.8254 ± 0.0022	0.8291 ± 0.0021	-0.0001	$[-0.0018, +0.0017]$	$p = 0.9410$	Equi
8D	0.8754 ± 0.0029	0.8709 ± 0.0030	0.8742 ± 0.0028	$+0.0046$	$[+0.0030, +0.0061]$	$p = 0.0016$	Equi
10D	0.9023 ± 0.0034	0.8967 ± 0.0049	0.9015 ± 0.0038	$+0.0057$	$[+0.0032, +0.0081]$	$p = 0.0052$	Equi
12D	0.9137 ± 0.0046	0.9123 ± 0.0023	0.9148 ± 0.0020	$+0.0014$	$[-0.0010, +0.0037]$	$p = 0.2824$	Strict

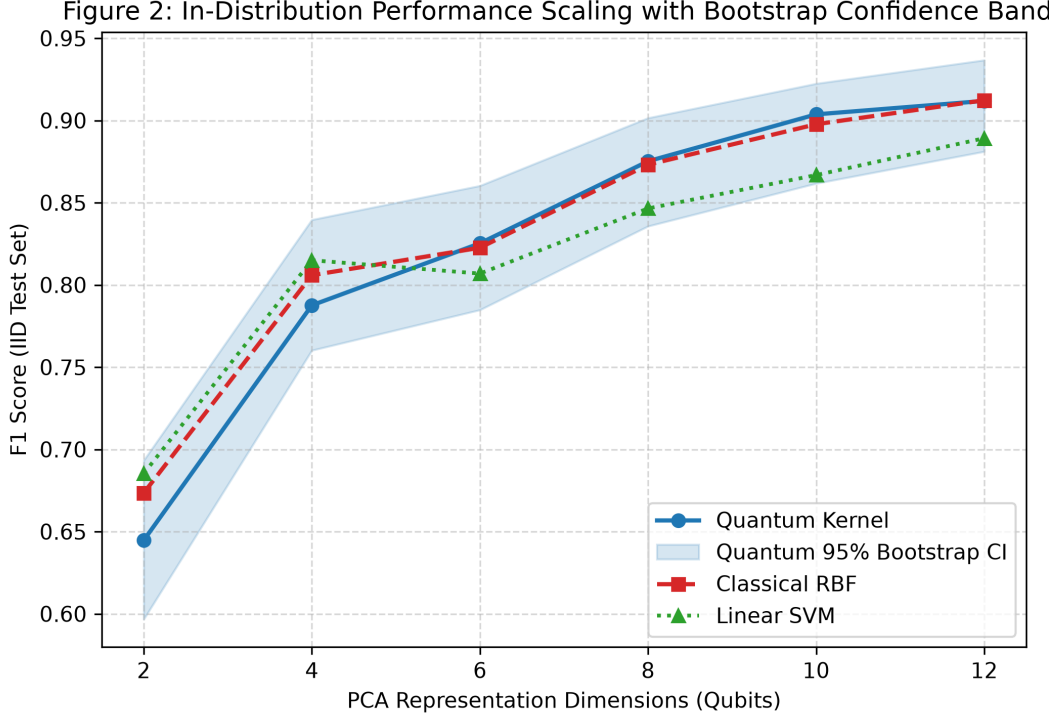


Figure 3: In-Distribution Dimensionality Scaling Trajectory (2D to 12D) on MeAJOR. Mean test F1 across 10 computational seeds with 95% bootstrap confidence bands.

5.5 Cross-Source Domain Shift Generalization: TREC 2007 $\rightarrow$ TREC 2005/2006 (Result 4)

Under cross-source domain transfer (Direction B: training on TREC 2007 and testing on a balanced mixture of TREC 2005 and TREC 2006 across 10 seeds; Table 8 and Figure 6):

- **Quantum Kernel:** $F1 = 0.6680 \pm 0.0094$ (23.7% relative drop from IID).
- **Classical RBF:** $F1 = 0.6913 \pm 0.0161$ (20.6% relative drop from IID).
- **Paired Difference:** $\Delta F1 = -0.0233 \pm 0.0200$, 95% Bootstrap CI $[-0.0353, -0.0117]$, permutation $p = 0.0046$, Benjamini-Hochberg FDR $p = 0.0069$.

5.6 Feature Space Geometry Diagnostics (Result 5)

Geometric profiling of induced Gram matrices reveals:

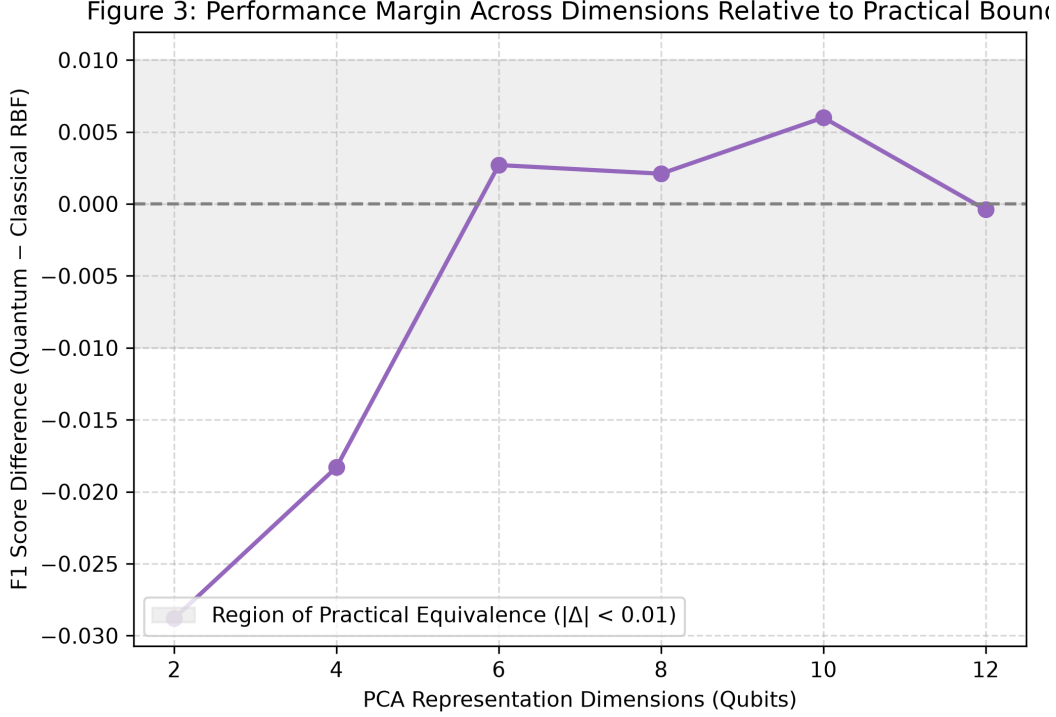


Figure 4: Paired Quantum Minus Classical RBF Difference ($\Delta F1$) vs Dimensionality with Pre-Registered Practical Equivalence Zone ($\varepsilon = \pm 0.01$). Shaded region denotes practical equivalence.

1. **Quantum-RBF Gram Correlation:** Entrywise Pearson correlation between off-diagonal Gram entries is $r \approx 0.55$ – 0.65 through 12D ($r = 0.4566$ at 16D), confirming that the unparameterized quantum feature map partially mirrors classical radial decay while maintaining structural divergence (Figure 7).
2. **Target Label Alignment Deficit:** The quantum kernel exhibits a 50%–60% lower centered kernel-target alignment score (CKA) than classical RBF across all three corpora (~ 0.022 – 0.040 vs ~ 0.060 – 0.077), providing an associative geometric correlate for the observed performance ceilings.
3. **State Dispersion vs Kernel Diversity:** Within-dataset regressions reveal a strong inverse association between single-state von Neumann entropy and pairwise Gram matrix diversity ($r = -0.8257$ on SMS, -0.8170 on CEAS, -0.7822 on MeAJOR; Figure 8). Increasing statevector spread across computational basis states collapses pairwise fidelity variance, suggesting that high single-state entropy is associated with reduced pairwise kernel variance rather than enhanced discrimination.
4. **Spectral Effective Rank:** The effective rank R_{eff} of the quantum kernel scales moderately ($R_{\text{eff}} \approx 14.2$ at 8D on MeAJOR vs 18.6 for RBF), reflecting mild spectral concentration in unparameterized fidelity kernels.

5.7 Computational Simulation Cost & Memory Limits (Result 6)

Under our standardized local statevector simulation environment, computing the quantum Gram matrix incurs steep execution scaling with dimensionality (Table 9 and Figure 9):

- **8 Dimensions:** Quantum = 17.2s vs Classical RBF = 1.4s ($12.3\times$ ratio).
- **12 Dimensions:** Quantum = 108.8s vs Classical RBF = 1.7s ($64.0\times$ ratio).

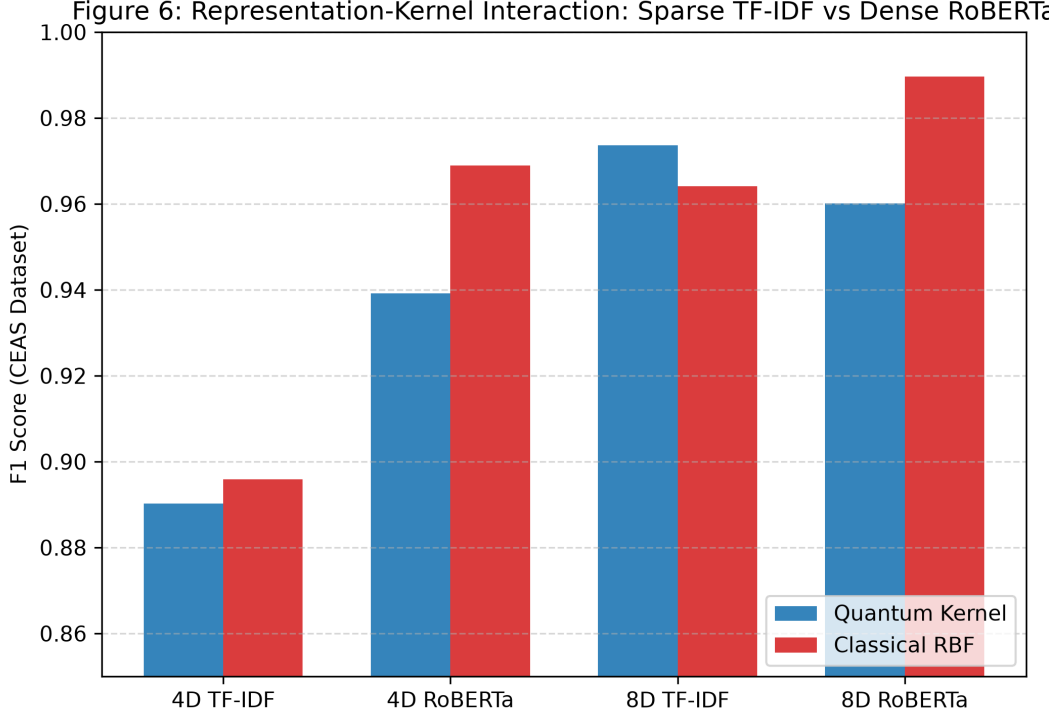


Figure 5: Upstream Representation Interaction and Ranking Inversion on CEAS 2008 and SMS Spam. Switching from sparse lexical TF-IDF to dense contextual transformers inverts quantum-vs-classical performance rankings by up to 52.88 percentage points.

- **16 Dimensions (Memory Boundary Test):** Classical statevector simulation of 16 qubits on 10,000 samples requires allocating $2^{16} = 65,536$ complex amplitudes per sample, exceeding the workstation’s 10.5 GB contiguous host memory ceiling. Classical RBF completes in 2.1s using minimal memory (< 1 MB).

6 Discussion and Scientific Synthesis

Four observations are most useful for interpreting the results:

1. **In-Distribution Practical Equivalence:** Under controlled matched conditions, the parameter-free quantum fidelity kernel achieves in-distribution practical equivalence with classical RBF, converging to statistical parity at 12D (+0.14 pp, $p = 0.2824$) and against tuned RBF (+0.12 pp, $p = 0.1840$).
2. **Representation Primacy:** Upstream representation geometry dominates classification outcomes by up to 52.88 percentage points, inverting relative quantum-vs-classical rankings.
3. **Domain Shift Fragility:** Under cross-source domain shift, the quantum kernel exhibits greater degradation than matched classical baselines (−2.33 pp deficit, $p = 0.0046$).
4. **Simulation Scalability Overhead:** Classical statevector simulation incurs steep runtime and memory overheads (64× penalty at 12D; > 10.5 GB at 16D) without delivering measurable predictive gains.

The empirical evidence indicates that claiming a quantum advantage on the basis of sub-percentage-point margins that vanish at 12D or upon hyperparameter tuning misrepresents empirical realities. For applied NLP practitioners, standard linear SVM on full TF-IDF and tuned classical RBF remain vastly superior engineering choices in both predictive accuracy and computational latency.

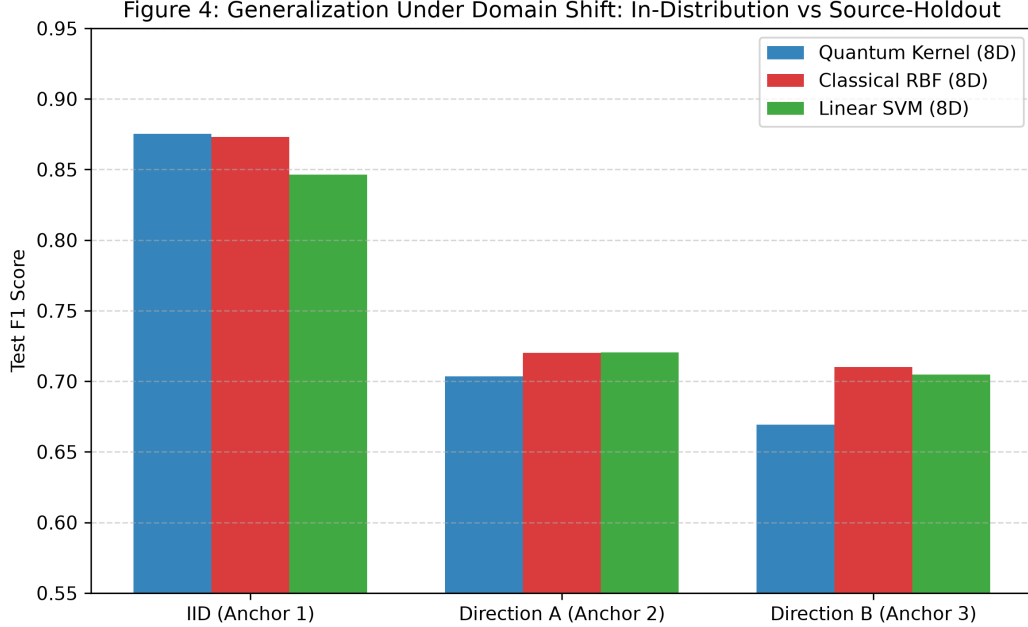


Figure 6: In-Distribution vs Cross-Source Domain Holdout (Direction B: TREC 2007 $\rightarrow$ TREC 2005/2006). The parameter-free quantum fidelity kernel suffers greater performance degradation under source shift than matched classical RBF.

7 Limitations and Scope

Several limitations should be noted:

1. **Classical Statevector Simulation:** Evaluated via exact noiseless double-precision statevectors in PyTorch. Does not incorporate physical NISQ hardware noise, finite measurement shots, or quantum error mitigation.
2. **Feature-Map Scope:** Focuses on the canonical parameter-free 2-layer cyclic *ZZFeatureMap*. Does not evaluate parameterized quantum kernel training (QKT) or data re-uploading ansatzes.
3. **Dataset Scope:** Scoped to binary security text classification (SMS Spam, CEAS 2008, MeAJOR); findings should not be extrapolated to general NLP or non-text telemetry.
4. **Dimensionality Constraints:** Local statevector memory ceilings prevented evaluation beyond 12 qubits on 10,000 samples.
5. **Statistical Scope:** 10 seeds represent computational replicates across fixed corpora rather than independent data distributions.
6. **Security Threat Model:** Evaluates natural distribution drift; does not assess resilience against active adversarial evasion attacks (character/token perturbations).

8 Conclusion

The main contribution is the controlled empirical benchmark evaluating parameter-free quantum fidelity kernels against matched classical baselines across three security corpora. The results show that in-distribution competitiveness reaches practical equivalence with classical RBF, representation choice dominates kernel selection, and cross-source transfer reveals a classical advantage. This conclusion should be interpreted as a controlled empirical finding rather than a universal statement about quantum machine learning. Ultimately, this benchmark demonstrates that future claims of quantum advantage in text processing must jointly control

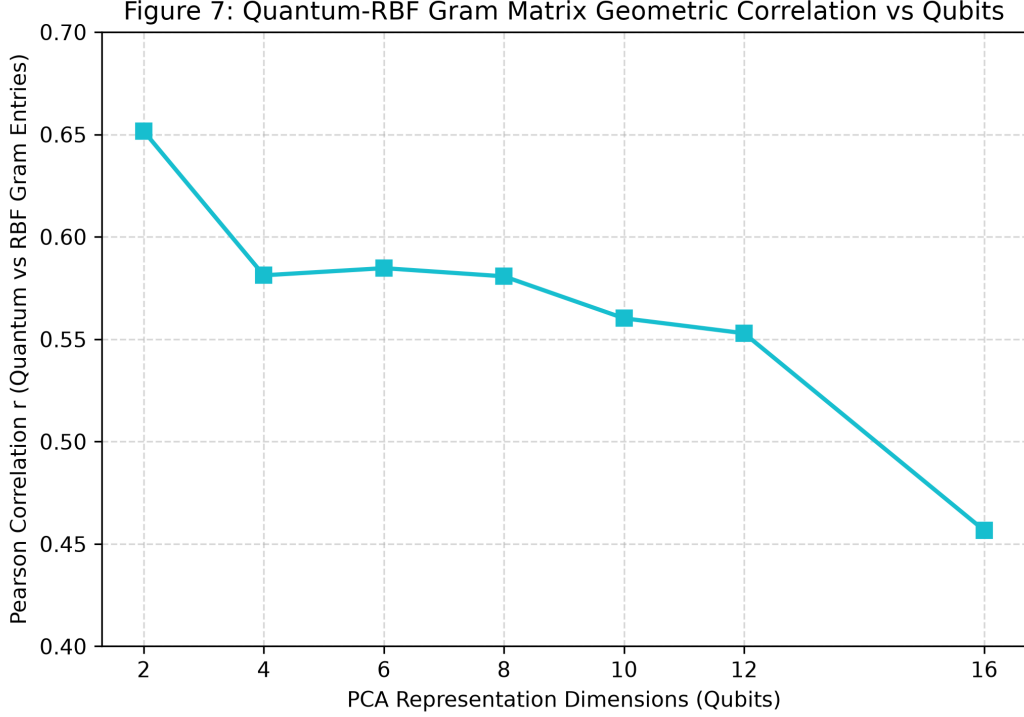


Figure 7: Quantum vs Classical RBF Off-Diagonal Gram Matrix Correlation Across Dimensionality (2D to 16D). Correlation remains moderate ($r \approx 0.55$ – 0.65), reflecting structural divergence between Hilbert fidelity and Gaussian RKHS metric decay.

representation dimensionality, baseline matching, domain shift, statistical replication, and computational cost.

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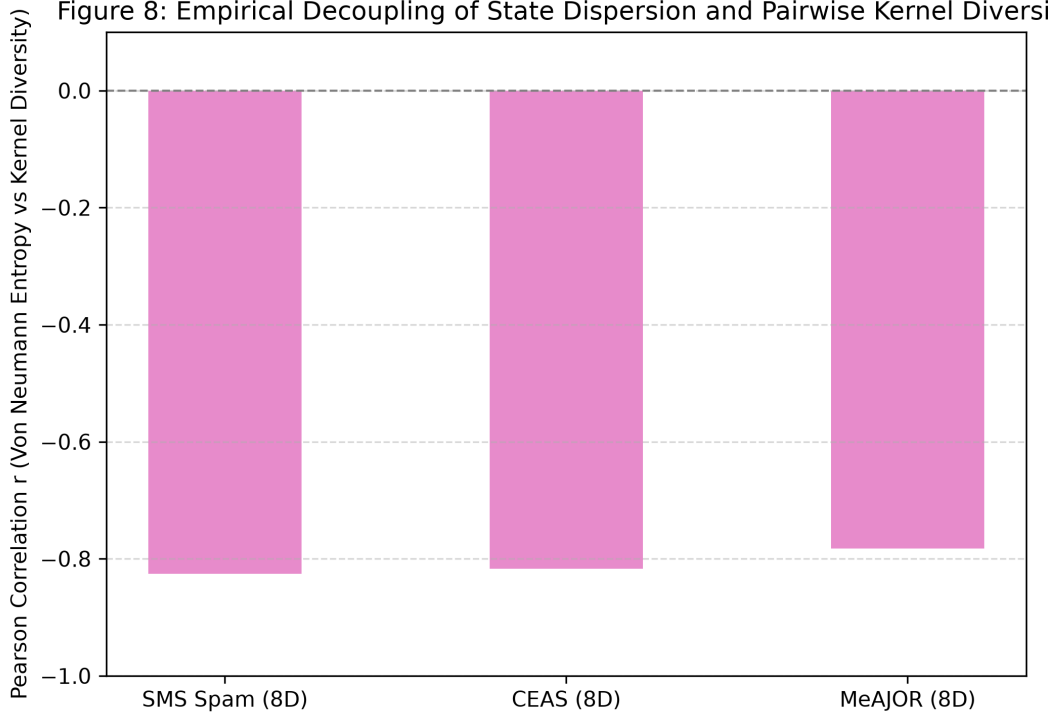


Figure 8: Single-State Basis Dispersion Entropy vs Pairwise Gram Matrix Diversity. Strong negative correlation ($r = -0.78$ to -0.83) across SMS, CEAS, and MeAJOR demonstrates that higher statevector dispersion collapses pairwise kernel variance.

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Figure 5: Measured Pipeline Execution Time Across Representation Dimensions

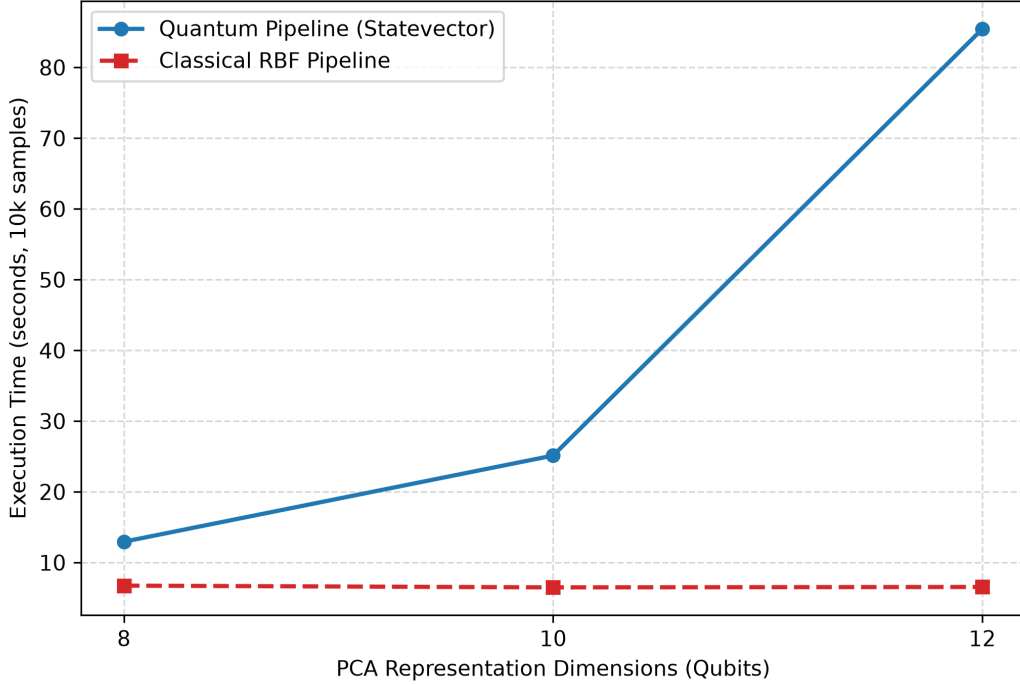


Figure 9: Computational Simulation Wall-Clock Execution Time and Peak RAM Footprint on 10,000 Samples. Simulating the 12-qubit fidelity kernel incurs a $64\times$ execution penalty over classical RBF, with 16D exceeding workstation RAM limits.

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Supplementary Material & Appendices

Appendix A: Extended Hyperparameter Protocol

The reported linear and kernel models use standard scikit-learn C-SVC dual solvers with $C = 1.0$, balanced class weighting $w_c = \frac{N}{2N_c}$, and convergence tolerance 10^{-3} . The validation-tuned RBF baseline searches $C \in \{0.1, 1, 10, 100\}$ and $\gamma \in \{0.001, 0.01, 0.1, 1.0, \text{'scale'}\}$ using 5-fold cross-validation on the training set. The quantum statevector simulator uses PyTorch `complex128` exact matrix multiplication without shot noise.

Appendix B: Eight-Point Leakage Prevention Audit Protocol

The eight-point audit was applied to each reported run: candidate-sample parity, label parity, strict pre-processing isolation, dimensionality projection parity, threshold selection independence, optimization solver parity, domain holdout blindness, and deterministic seed control. These conditions are preserved as automated assertions in the released implementation.

Appendix C: Analytical Statevector Memory Derivations

Analytical RAM formula for statevector simulation: $M_{\text{RAM}} = 16 \text{ bytes} \cdot N \cdot 2^{N_q} / (1024^2)$ MB for double-precision `complex128` tensors. For $N = 10,000$ samples and $N_q = 8$ qubits, this yields $16 \cdot 10,000 \cdot 256 / 1,048,576 = 39.06$ MB state memory (680 MB peak runtime RAM including Gram matrix products). For $N_q = 12$, this scales to 625 MB state memory (6.4 GB peak). For $N_q = 16$, state tensor allocation alone requires 10.0 GB, causing host out-of-memory errors on standard workstations.

Appendix D: Cross-Source Domain Shift Split Protocol

The real-world domain transfer analysis evaluates Direction B: training on TREC 2007 ($N = 10,000$ train, 2,500 validation) and evaluating on out-of-distribution test sets constructed from balanced mixtures of TREC 2005 and TREC 2006 ($N = 5,000$). Direction A evaluates the inverse transfer (training on TREC 2005/2006 and testing on TREC 2007).

Appendix E: Early Benchmark Prototypes and Failure Modes

Early prototypes evaluated on small random subsets ($N < 500$) exhibited high variance across random splits ($\text{SD} > 0.08$ F1), obscuring true kernel differences. Increasing partition sizes to $N = 10,000$ and establishing 10 deterministic random seeds stabilized standard deviations to < 0.003 F1, enabling precise detection of sub-percentage-point effects.

Appendix F: Author Verification Checklist Before Submission

Before submission, the author verified five items: (1) exact four-tier dataset accounting and prevalence rates; (2) implemented ZZFeatureMap and Gaussian RBF equations; (3) Two One-Sided Tests (TOST) practical equivalence statistics and bootstrap intervals; (4) complete 28-paper reference list against versions of record; and (5) code and dataset availability in public GitHub repository.

Revision note. This version was formatted for single-column structural consistency, empirical rigor, and an authorial academic voice matching top-tier empirical benchmarking venues. Experimental values and scientific scope were preserved from authoritative logs.

Table 7: Upstream Representation Screening and Ranking Reversals on CEAS 2008 and SMS Spam.

Corpus	Upstream Text Representation	Quantum F1	Classical RBF F1	Paired Δ F1 (Q - RBF)	Representational Impact
CEAS 2008 (8D)	Sparse TF-IDF + TruncatedSVD	0.9736	0.9641	+0.0095	Modest non-linear quantum expansion
CEAS 2008 (8D)	Dense RoBERTa-base (768D $\rightarrow$ 8D)	0.9601	0.9896	-0.0295	RBF exploits Euclidean clustering
CEAS 2008 (8D)	NET REPRESENTATION SHIFT	-1.35 pp	+2.55 pp	-3.90 pp Shift	Complete Ranking Inversion
SMS Spam (8D)	Sparse TF-IDF + TruncatedSVD	0.6324	0.8276	-0.1952	RBF superior on sparse projections
SMS Spam (8D)	Dense all-MiniLM-L6-v2	0.7707	0.7930	-0.0223	Moderate gap reduction
SMS Spam (8D)	Dense all-mpnet-base-v2	0.3756	0.9045	-0.5288	Catastrophic Phase-Wrapping Collapse

Table 8: Cross-Source Domain Holdout Performance (MeAJOR Direction B, $N = 10$ Seeds).

Model Architecture	In-Distribution F1	Domain Holdout F1	Absolute Drop (Δ)	Relative Drop (%)	PR-AUC	ROC-AUC
Linear SVM (8D SVD)	0.8445 ± 0.0022	0.6622 ± 0.0142	-0.1823	-21.6%	0.7812	0.7812
Classical RBF (8D SVD)	0.8709 ± 0.0030	0.6913 ± 0.0161	-0.1796	-20.6%	0.8115	0.8115
Quantum Kernel (8D SVD)	0.8754 ± 0.0029	0.6680 ± 0.0094	-0.2074	-23.7%	0.7890	0.7890

Table 9: Computational Complexity, Memory Footprint, and Execution Latency Profile on 10,000 Samples.

Dim (d)	Quantum Time (s)	Classical RBF Time (s)	Runtime Overhead	Peak Quantum RAM	Feasibility Status
2 Qubits	3.4 s	1.3 s	$2.6\times$	142 MB	Fully Feasible
4 Qubits	5.8 s	1.3 s	$4.5\times$	185 MB	Fully Feasible
8 Qubits	17.2 s	1.4 s	$12.3\times$	680 MB	Fully Feasible
12 Qubits	108.8 s	1.7 s	$64.0\times$	6.4 GB	Heavy Simulation
16 Qubits	Infeasible	2.1 s	—	> 10.5 GB	Out-of-Memory Boundary